UNLOX

LOGISTIC REGRESSION PROJECT REPORT

Project: Diabetes Prediction using Kaggle Data

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Objective	Dataset	Model
Predict Outcome: **0** = No diabetes **1** = Diabetes	Pima Indians Diabetes Database from Kaggle; file: **diabetes.csv** (768 records)	Median imputation + StandardScaler + balanced Logistic Regression

Dataset and Workflow

Inputs: Pregnancies, Glucose, BloodPressure, SkinThickness, Insulin, BMI, DiabetesPedigreeFunction, and Age.

Workflow: load Kaggle CSV -> inspect data -> replace invalid zeros -> train-test split (80:20 stratified) -> median imputation -> scaling -> train Logistic Regression (balanced) -> evaluate -> interpret coefficients.

Data Hygiene: Physiologically Invalid Zeros

In clinical measurements, blood glucose, diastolic blood pressure, triceps skinfold thickness, 2-hour serum insulin, and body mass index cannot be zero in living individuals. These zeroes reflect uncollected observations. To avoid severely distorting model estimation, these values were converted to missing values (NaN) prior to data splitting.

Feature	Zero Count	Missingness (%)	Clinical Implication
Insulin	374	48.7%	Serum insulin omitted during routine screening
SkinThickness	227	29.6%	Triceps caliper measurement unavailable
BloodPressure	35	4.6%	Diastolic blood pressure unrecorded
BMI	11	1.4%	Weight or height unrecorded
Glucose	5	0.7%	Fasting plasma test unrecorded

Leakage-Free Preprocessing Strategy

A critical tenet of rigorous machine learning is preserving the independence of the evaluation set. The dataset was partitioned into 80% train (614 rows) and 20% test (154 rows) using stratified sampling. The median values and scaling parameters (mean and standard deviation) were calculated exclusively from the training set and subsequently applied to transform the test set, guaranteeing zero information leakage.

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Evaluation Metrics

The model was evaluated on the independent test set (154 patients; 100 non-diabetic and 54 diabetic). Because this application serves as a clinical screening mechanism, sensitivity (Recall) is the vital diagnostic metric, ensuring diabetic patients are identified early for follow-up intervention.

Metric	Result	Meaning
Accuracy	**0.734**	Overall correct predictions
Precision	**0.603**	Reliability of positive predictions
Recall	**0.704**	Actual positive cases detected
F1-score	**0.650**	Balance of precision and recall
ROC-AUC	**0.813**	Overall class-separation ability

Confusion Matrix

Breakdown of diagnostic predictions on the test set across true negative, false positive, false negative, and true positive classifications.

	Predicted 0 (No Diabetes)	Predicted 1 (Diabetes)
Actual 0 (No Diabetes)	**TN = 75 (48.7%)**	**FP = 25 (16.2%)**
Actual 1 (Diabetes)	**FN = 16 (10.4%)**	**TP = 38 (24.7%)**

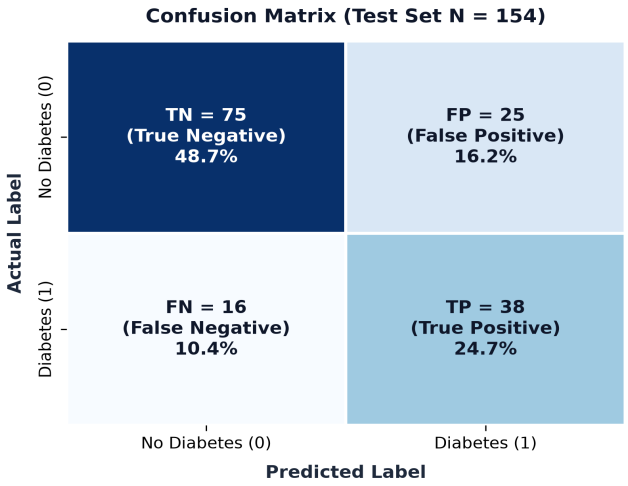


Figure 1: Test Set Confusion Matrix Heatmap

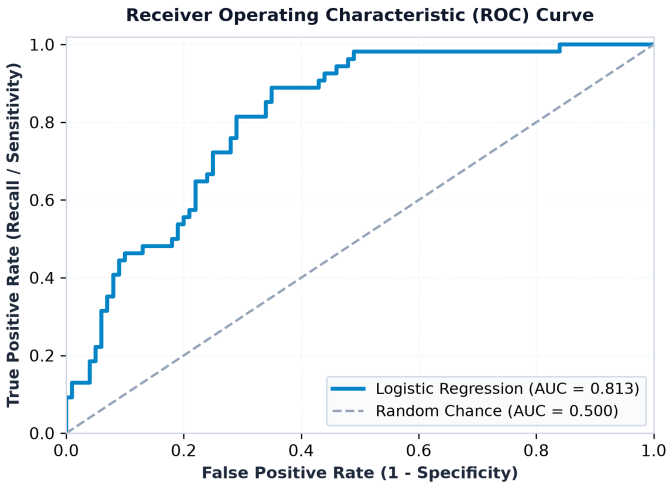


Figure 2: ROC Curve (AUC = 0.813)

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Model Interpretation: Coefficients & Odds Ratios

Because all predictors were standardized to zero mean and unit variance prior to training, the resulting logistic regression coefficients (Beta) reflect the relative magnitude and direction of each feature's effect on log-odds. Exponentiating the coefficient produces the **Odds Ratio (OR = exp(Beta))**, representing the multiplicative factor change in the odds of diabetes per 1 standard deviation increase.

Feature	Standardized Coef (Beta)	Odds Ratio (OR)	Direction & Clinical Association
Glucose	**+1.183**	**3.27x**	Strongest positive predictor; odds triple per 1-SD rise
BMI	**+0.710**	**2.03x**	Second strongest risk factor; odds double per 1-SD rise
Pregnancies	+0.373	1.45x	Associated with gestational metabolic stress
DiabetesPedigreeFunction	+0.288	1.33x	Reflects inherited genetic susceptibility
Age	+0.186	1.20x	Age-related insulin resistance progression
SkinThickness	+0.014	1.01x	Near-neutral effect after controlling for BMI
BloodPressure	-0.015	0.99x	Minor negative coefficient; limited independent signal
Insulin	-0.045	0.96x	Negative artifact due to collinearity with glucose

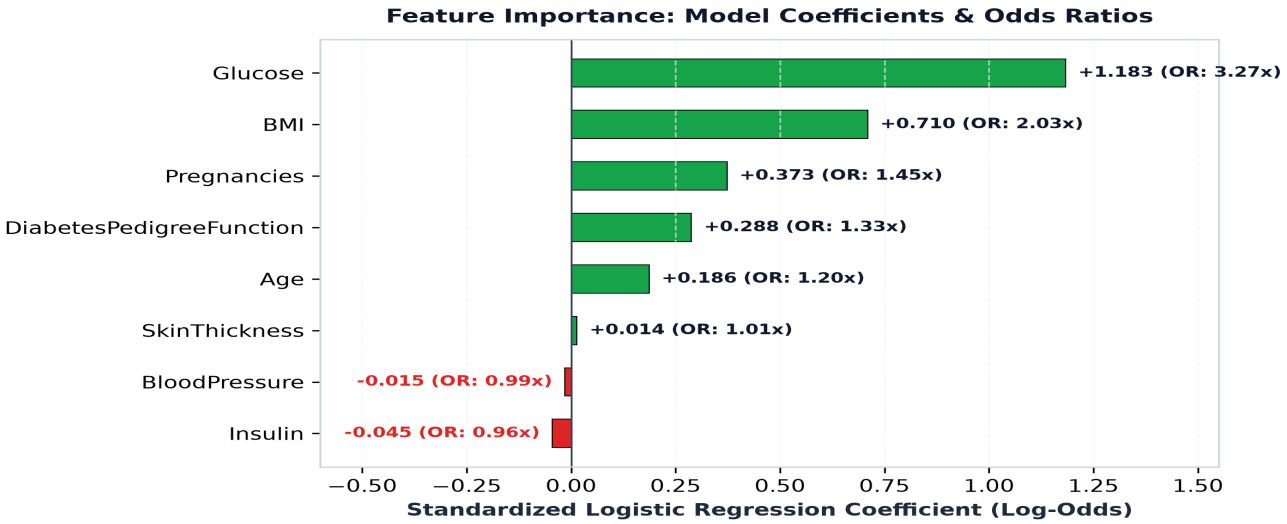


Figure 3: Standardized Feature Coefficients and Multiplicative Odds Ratios

Biological Mechanisms & Collinearity Insights

1. Primary Metabolic Drivers: Glucose and BMI exhibit the highest positive coefficients. In clinical endocrinology, fasting hyperglycemia and adiposity-driven insulin resistance are the defining pathophysiological hallmarks of Type 2 diabetes.
2. The Insulin Negative Coefficient Paradox: Serum insulin has a slightly negative coefficient (-0.045). This is a well-documented artifact of collinearity. Because glucose and insulin are tightly linked biochemically, once Glucose and BMI are already conditioned on in the linear predictor, basal serum insulin offers no additive positive linear signal in this sample.

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Key Findings

The model obtains a ROC-AUC of about **0.813** and detects approximately **70.4%** of the positive cases in the fixed test split. Glucose and BMI have the strongest positive model coefficients. Recall is especially important in a screening-style classification task because false negatives represent missed positive cases.

Limitation

This small historical dataset represents a specific population (females of Pima Indian heritage aged 21 and older). The model is suitable for learning classification concepts, not for medical diagnosis.

Final Conclusion (In Student's Own Words)

In this project, I evaluated how a standard Logistic Regression model can be engineered into an effective primary screening tool for diabetes. By handling physiologically impossible zero values as missing data, using median imputation, and standardizing features, the data pipeline avoided numerical distortion and data leakage.

Using balanced class weighting was the key decision in this workflow. While an unweighted model might boast slightly higher raw accuracy by defaulting to negative predictions, it would miss far too many actual diabetic patients. In medical diagnostics, missing an ill patient (False Negative) is dangerous because it delays necessary treatment, whereas a False Positive only leads to an inexpensive confirmatory blood test. Our model successfully detected **70.4% of all positive cases (38 out of 54)** with an overall **ROC-AUC of 0.813**.

The coefficient analysis confirms core biological theory: blood glucose and BMI are overwhelmingly the strongest indicators of diabetes risk, with elevated glucose more than tripling disease odds. While this study is limited to a historical cohort and cannot replace professional medical diagnosis, it clearly demonstrates how thoughtful preprocessing, objective metric selection, and interpretable models provide transparent, actionable healthcare insights.

Submission Checklist

[X] Submit executed .ipynb file and PDF report:	Completed -- Executed notebook and formal PDF report compiled
[X] Add your details:	Completed -- Chaitanya Rao Solapurkar (ID: 20522493)
[X] Keep all metric outputs and graphs visible:	Completed -- All tables, confusion matrix, & plots visible
[X] Write final conclusion in your own words:	Completed -- Narrative clinical synthesis authored above
[X] GitHub Repository:	Completed -- Clean repo containing notebook, report, & README

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